Linear Algebra — Lecture 9–10: Eigenvalues and Diagonalization

CE2005 Linear Algebra, Fall 2026

Eigenvalues and Eigenvectors

Definition. Let A be an $n \times n$ matrix. A nonzero vector $\mathbf{x}$ is an *eigenvector* of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. The scalar λ is called an *eigenvalue* of A.

- λ is an eigenvalue of A iff $(A - \lambda I)\,\mathbf{x} = \mathbf{0}$ has a nontrivial solution.
- The set of all solutions of $(A - \lambda I)\,\mathbf{x} = \mathbf{0}$ is the **eigenspace** of A corresponding to λ.

Theorem 1. The eigenvalues of a triangular matrix are the entries on its main diagonal.

Theorem 2. If $\mathbf{v}_1, ..., \mathbf{v}_r$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, ..., \lambda_r$, then $\{\mathbf{v}_1, ..., \mathbf{v}_r\}$ is linearly independent.

Note: 0 is an eigenvalue of A iff A is not invertible.

The Characteristic Equation

A scalar λ is an eigenvalue of A iff λ satisfies the characteristic equation

$$\det\ (\ A - \lambda I\ ) = 0.$$

Example. $A = \begin{pmatrix} 2 & 3 \\ 3 & -6 \end{pmatrix}$. $\det\ (\ A - \lambda I\ ) = (\ 2 - \lambda\ )\ (\ -6 - \lambda\ ) - 9 = \lambda^2 + 4\lambda - 21 = (\ \lambda - 3\ )\ (\ \lambda + 7\ )$. Eigenvalues: $3,\ -7$.

- **Algebraic multiplicity**: multiplicity of λ as a root of the characteristic polynomial.
- **Geometric multiplicity**: $\dim$ of the eigenspace. Always $1 \leq$ geo. mult. $\leq$ alg. mult.

Similarity. A and B are *similar* if $B = P^{-1}AP$ for some invertible P. Similar matrices have the same characteristic polynomial (hence same eigenvalues). The converse is FALSE.

Useful facts: $\det A = \prod \lambda_i,\ \operatorname{tr} A = \sum \lambda_i$.

Diagonalization

A is **diagonalizable** if $A = PDP^{-1}$ with D diagonal.

The Diagonalization Theorem. An $n \times n$ matrix A is diagonalizable iff A has n linearly independent eigenvectors. In that case the columns of P are the eigenvectors and the diagonal entries of D are the corresponding eigenvalues (same order!).

Theorem. An $n \times n$ matrix with n distinct eigenvalues is diagonalizable. (Sufficient, NOT necessary.)

Theorem. A is diagonalizable iff for each eigenvalue, geometric multiplicity = algebraic multiplicity.

Application: $A^k = PD^kP^{-1}$.

Example. $A = \begin{pmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{pmatrix}$ has eigenvalues 1 and -2 (alg. mult. 2). Eigenspace of -2 is 2-dimensional, so A is diagonalizable.

Non-example. $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$: eigenvalue 2 with alg. mult. 2, geo. mult. 1 — not diagonalizable.

Complex Eigenvalues (brief)

Real matrices may have complex eigenvalues, which come in conjugate pairs. Example: rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has eigenvalues $\pm i$.